

Predicted vs Observed Metastasis-Free Survival in Individuals With Uveal Melanoma

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IMPORTANCE Accuracy of the predicted metastasis-free survival (MFS) by a commercially available gene expression profiling (GEP) test is not known.

OBJECTIVE To compare the predicted MFS with the observed MFS in patients in this cohort and with those in published studies (published MFS, meta-analysis).

DESIGN, SETTING, AND PARTICIPANTS This cohort study included consecutive patients from the University of Iowa and Cleveland Clinic who were diagnosed with uveal melanoma who underwent prognostic fine-needle aspiration biopsy at the time of primary treatment. Patients were recruited from December 2012 to December 2020. The predicted MFS for patients was extracted from the GEP report. The observed MFS was defined as time to metastasis. Cox proportional hazards models were fit to identify tumor variables impacting MFS in patients with class 2 tumors. The overall estimate of the published MFS was obtained by performing meta-analysis of data from published series. Analysis took place in August 2021.

MAIN OUTCOMES AND MEASURES

RESULTS There were 92 patients from the University of Iowa and 255 patients from the Cleveland Clinic. The mean (SD) age at diagnosis was 59.4 (13.0) years. The median (IQR) follow-up interval was 38.0 (19.0-57.0) months. The observed MFS for patients with class 2 tumor in this cohort (3 years: 67% [95% CI, 59%-77%]; 5 years: 47% [95% CI, 37%-61%]) and in published studies (3 years: 62% [95% CI, 57%-66%]; 5 years: 40% [95% CI, 34%-46%]) were better than those predicted (50% and 28% for 3 and 5 years, respectively). Within patients with class 2 tumor, those with metastasis had larger tumors compared with nonmetastatic tumors (mean largest basal diameter difference, 1.7 [95% CI, 0.5-3.0] mm; $P = .01$; mean thickness ratio, 1.3 [95% CI, 1.04-1.5]; $P = .01$, respectively). An increasing tumor size was significantly associated with increased hazard ratio (1.16 [95% CI, 1.06-1.27]; $P < .001$) of metastasis.

CONCLUSIONS AND RELEVANCE These findings suggest the predicted MFS for metastatic tumors (class 2) appears to be worse than that observed here and reported by others. Incorporation of tumor size in the prediction model may enhance its accuracy. Adjuvant therapy trials may not be able to rely on predicted MFS to calculate efficacy with a high degree of confidence.

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 Multimedia

 Supplemental content

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Since the initial report by Prescher et al¹ in 1992 that suggested correlation between cytogenetic changes and prognosis of uveal melanoma, several groups have reported on prognostication methods, prognostic classifications, and development of prognostic tests with direct clinical applications.^{2,3} Although various molecular techniques have been explored, at present only 2 prognostication tests, multiplex ligation-dependent probe amplification (Impact Genetics)^{4,5} and gene expression profiling (GEP; DecisionDx-UM; Castle Biosciences),⁶ are commercially available in the US. Since 2012, we (A.D.S., E.M.B., and H.C.B.) have offered GEP testing to all patients with uveal melanoma integral to their management plan for tailoring systemic surveillance and consideration for enrollment into adjuvant therapy trials.⁷

Castle Biosciences reports tumors by their molecular signature as class 1A (low risk), class 1B (intermediate risk), or class 2 (high risk).⁶ The predicted survival is reported as percent metastasis free at 3 and 5 years of 98% and 98%, 93% and 79%, and 50% and 28% for class 1A, class 1B, or class 2 tumors, respectively. Of note, the predicted metastasis-free survival (MFS) lacks confidence intervals, a measure of accuracy (the narrower the range, the more robust the prediction), reproducibility, and its application in larger sample population. Because confidence intervals quantify the uncertainty of an estimate by providing the magnitude of the estimate relative to the magnitude of random chance, they allow assessment of statistical significance.⁸

Following the publication of the 2012 Collaborative Ocular Oncology Group (COOG) study results,⁶ several single-institutional experiences have been reported.⁹⁻¹⁴ Yet, there is lack of a formal comparative analysis of survival outcomes in a community setting scenario. Therefore, we undertook this analysis to compare the predicted MFS with the observed MFS in patients and with those in published studies (published MFS, meta-analysis). We also explored the association of tumor variables, including tumor size, with the MFS in patients predicted to have high risk of metastasis (class 2).

Methods

Institutional review board approval was obtained from both the University of Iowa and the Cleveland Clinic. This research adhered to the tenets of the Declaration of Helsinki.¹⁵ Informed consent was not required from participants, and participants did not receive any compensation or incentives to participate. Records of consecutive patients treated for uveal melanoma with episcleral plaque brachytherapy at the University of Iowa and with episcleral plaque brachytherapy, enucleation, or resection at the Cleveland Clinic and who underwent fine-needle aspiration biopsy prognostic GEP testing were reviewed. The University of Iowa began offering GEP testing in December 2012 and the Cleveland Clinic in November 2013. Patients were recruited from December 2012 and December 2020. Patients were excluded if biopsy results were unavailable owing to technical failure. GEP tests were performed as part of standard of care and the results were provided by Castle Biosciences.

Key Points

Question Is the survival of patients with uveal melanoma predicted by gene expression profiling test accurate?

Findings In this cohort study, the observed metastasis-free survival at 3 and 5 years for patients with class 2 tumor in this cohort and in published studies (62% and 42%, respectively) were not different and potentially better than those predicted. Within patients with class 2 tumor, those with metastasis had larger tumors compared with nonmetastatic tumors; an increasing tumor size was associated with increased hazard ratio of metastasis.

Meaning This analysis raises questions about the accuracy of survival prediction of patients with uveal melanoma.

Patient age at diagnosis, tumor largest basal diameter (LBD) and thickness, tumor location, treatment modality (episcleral plaque brachytherapy, enucleation, other [transpupillary thermotherapy, resection, proton beam radiation]), date of treatment, biopsy technique, GEP class, predicted metastasis rate, date of last follow-up, presence of metastasis, and status at last follow-up (alive or deceased) was recorded for each patient. Data on race and ethnicity were not collected. The date of last follow-up was determined by review of the electronic medical record and was defined as the date of last follow-up with either the ocular oncology or medical oncology service. For patients who transferred care outside of either institution, the date of last follow-up with local clinicians was recorded where available. Each tumor was also classified using the 8th edition of the American Joint Committee on Cancer (AJCC) staging manual for uveal melanoma and by the Collaborative Ocular Melanoma Study (COMS) small, medium, and large criteria.^{16,17}

For patients who developed metastatic disease, the modality by which metastasis was detected and confirmed was recorded. These data were obtained by review of the electronic medical record and review of outside documents including imaging results and pathologic reports where available. For patients who were deceased, cause and date of death was determined by review of the electronic medical record, obituaries, notification by family or friends where applicable, and report by the cancer registry at either the University of Iowa or Cleveland Clinic. The means by which these data were obtained (electronic medical record, cancer registry, family or friend report, obituary) were recorded for each patient. Mortality data were verified with the records of the oncology registry at the University of Iowa and the tumor registry at the Cleveland Clinic Taussig Cancer Institute.

The predicted MFS data for each patient were extracted from the GEP report. The observed MFS was defined as time to metastasis diagnosis. The last follow-up date was used for censoring those without these events. Categorical variables were described using frequencies and percentages, while continuous measures were described with means and standard deviations or medians and interquartile ranges. Comparisons were made using Pearson χ^2 tests or Fisher exact tests for categorical variables and *t* tests or Kruskal-Wallis tests for continuous measures. Cox proportional hazards models were fit for MFS to assess whether tumor variables influenced MFS in

Table 1. Overall Summary Data

Characteristic	No. (%)		Effect (95% CI)	P value
	University of Iowa (n = 92)	Cleveland Clinic (n = 255)		
Age at diagnosis, mean (SD), y	60.2 (12.0)	59.1 (13.4)	-1.1 (-4.2 to 2.1)	.50 ^a
Eye				
Left	47 (51.1)	127 (49.8)	1 [Reference]	.83 ^b
Right	45 (48.9)	128 (50.2)	1.05 (0.65 to 1.7)	
Location				
Choroid only	72 (78.3)	200 (78.4)	1 [Reference]	.02 ^c
Ciliary body ± choroid	20 (21.7)	38 (14.9)	0.68 (0.37 to 1.3)	.22 ^b
Iris	0	17 (6.7)	NE	NA
COMS size				
Small	7 (7.6)	77 (30.2)	1.01 (0.71 to 1.4)	.97 ^d
Medium	79 (85.9)	103 (40.4)		
Large	6 (6.5)	75 (29.4)		
AJCC stage				
I	16 (17.4)	82 (32.2)	1.4 (1.07 to 1.9)	.01 ^d
II	67 (72.8)	65 (25.5)		
III	9 (9.8)	84 (32.9)		
IV	0	24 (9.4)		
Tumor size, mm ^e				
LBD, mean (SD)	12.5 (2.2)	12.3 (4.0)	-0.2 (-1.1 to 0.7)	.67 ^b
Height, median (IQR)	3.5 (2.8 to 4.7)	4.5 (2.7 to 8.0)	1.2 (1.1 to 1.4)	.002 ^f
Treatment				
Enucleation	0	68 (26.7)	1 [Reference]	<.001 ^g
Episcleral brachytherapy	92 (100.0)	182 (71.4)	NE	NA
Other ^g	0	5 (2.0)	NE	NA
GEP class				
1A	33 (35.9)	117 (45.9)	0.88 (0.67 to 1.2)	.36 ^c
1B	27 (29.3)	49 (19.2)		
2	32 (34.8)	89 (34.9)		
All deaths	18 (19.6)	38 (14.9)	0.72 (0.39 to 1.3)	.30 ^b
Metastasis	14 (15.2)	34 (13.3)	0.86 (0.44 to 1.7)	.65 ^b
Follow-up, median (IQR), mo	49.5 (37.0 to 67.0)	33.0 (17.0 to 52.0)	0.7 (0.6 to 0.8)	<.001 ^f

Abbreviations: AJCC, American Joint Committee on Cancer; COMS, Collaborative Ocular Melanoma Study; GEP, gene expression profiling; LBD, largest basal diameter; NA, not applicable; NE, not estimable.

^a Statistics presented as mean (SD) with mean difference and P value from linear regression.

^b Statistics presented as No. (%) with odds ratios from logistic regression.

^c NE odds ratios are listed when cell sizes with 0 patients existed, and overall Pearson χ^2 tests are shown.

^d Statistics presented as No. (%) with odds ratios from proportional odds regression.

^e Data not available for all individuals. Missing values: tumor size: LBD = 2; tumor size: height = 4.

^f Statistics presented as median (IQR) with mean ratios.

^g Other treatment includes excision, transpupillary thermotherapy, proton beam radiation therapy.

patients with class 2 tumors. Effect estimates for the group comparisons were included and used odds ratios for categorical and ordinal outcomes, mean differences for normally distributed continuous measures, and mean ratios for non-normal continuous measures that were log-transformed to better align with normality and then back-transformed for presentation. The published MFS was obtained by performing a meta-analysis of data from published series. Studies were identified by searching in PubMed using the search terms *gene expression profiling* and *uveal melanoma/choroidal melanoma, prognostication*, and *uveal/choroidal melanoma*. Articles using the GEP prognostication test were identified. For inclusion, articles must have reported both the 3-year and 5-year MFS for the entire cohort of patients. The approach described by Combes et al¹⁸ was used with the metaSurvival package in R version 4.0 (R Foundation). Briefly, the digitized package was used to estimate MFS at 3-month intervals through 5 years, and the number at risk at each time point was estimated either directly from the plots or by adapting the approach by Parmar et al¹⁹ to fit a quadratic function for time at risk based

on the minimum, maximum, and median follow-up times.¹⁹ P values were 2-sided and not adjusted for multiple comparisons. Analysis was performed using SAS statistical software version 9.4 (SAS Institute) and R version 4.0 (R Foundation). Analysis took place in August 2021.

Results

General Demographic Data

There were 92 patients from the University of Iowa and 255 patients from the Cleveland Clinic. The mean (SD) age at diagnosis was 59.4 (13.0) years. There were 173 right eyes and 174 left eyes. The mean (SD) tumor LBD was 12.4 (3.6) mm and the median (IQR) thickness was 4.0 (2.7-7.2) mm. A total of 272 tumors (78%) were choroidal, 58 (17%) involved the ciliary body ± choroid, and 17 (5%) were confined to the iris. By COMS classification, 82 tumors (24%) were small, 181 tumors (53%) were medium, and 81 tumors (23%) were large.²⁰ By AJCC staging, 98 tumors (28%) were stage I, 132 (38%) were stage II, 93

Table 2. Clinical Features Associated With Metastasis in Patients With Gene Expression Profiling Prognostic Class 2 Uveal Melanoma

Characteristic	No. (%)		Effect (95% CI)	P value
	No metastasis (n = 81)	Metastasis (n = 40)		
Eye				
Left	45 (55.6)	18 (45.0)	1 [Reference]	.27 ^a
Right	36 (44.4)	22 (55.0)	1.5 (0.71-3.3)	
Location				
Choroid only	60 (74.1)	32 (80.0)	1 [Reference]	.65 ^b
Ciliary body ± choroid	20 (24.7)	8 (20.0)	0.75 (0.30-1.9)	.54 ^a
Iris	1 (1.2)	0	NE	NA
COMS				
Small	13 (16.0)	3 (7.5)	1.8 (0.99-3.5)	.06 ^c
Medium	48 (59.3)	21 (52.5)		
Large	20 (24.7)	16 (40.0)		
AJCC				
I	19 (23.5)	3 (7.5)	1.6 (1.02-2.5)	.04 ^c
II	29 (35.8)	15 (37.5)		
III	26 (32.1)	16 (40.0)		
IV	7 (8.6)	6 (15.0)		
Tumor size, mm				
LBD, mean (SD)	12.7 (3.3)	14.5 (3.1)	1.7 (0.5-3.0)	.01 ^d
Height, median (IQR)	4.0 (3.1-6.7)	5.9 (3.6-9.3)	1.3 (1.04-1.5)	.01 ^e
Treatment				
Enucleation	15 (18.5)	13 (32.5)	1 [Reference]	.16 ^b
Episcleral brachytherapy	64 (79.0)	27 (67.5)	0.49 (0.20-1.2)	.10 ^a
Other ^f	2 (2.5)	0	NE	NA
Follow-up, median (IQR), mo	36.0 (18.0-58.0)	39.5 (28.5-55.5)	1.2 (0.94-1.6)	.12 ^e

Abbreviations: AJCC, American Joint Committee on Cancer; COMS, Collaborative Ocular Melanoma Study; LBD, largest basal diameter; NA, not applicable; NE, not estimable.

^a Statistics presented as No. (%) with odds ratios from logistic regression.

^b NE odds ratios are listed when cell sizes with 0 patients existed, and overall Pearson χ^2 tests are shown.

^c Statistics presented as No. (%) with odds ratios from proportional odds regression.

^d Statistics presented as mean (SD) with mean difference and P value from linear regression.

^e Statistics presented as median (IQR) with mean ratios.

^f Other treatment includes excision, transpupillary thermotherapy, and proton beam radiation therapy.

(27%) were stage 3, and 24 (7%) were stage IV.²¹ Sixty eight eyes (20%) were treated with enucleation, 274 (79%) with I-125 episcleral plaque brachytherapy, and 5 (1%) with other methods (excision, transpupillary thermotherapy, proton beam radiation therapy) (Table 1).

GEP Class

There were 150 class 1A tumors (43%), 76 class 1B tumors (22%), and 121 class 2 tumors (35%). Throughout a median (IQR) follow-up interval of 38.0 (19.0-57.0) months, 48 patients developed metastatic disease. Metastasis was diagnosed in 48 patients (5 with class 1A, 3 with class 1B, and 40 with class 2 tumors) by surveillance imaging (magnetic resonance imaging, computed tomography, or right upper quadrant ultrasonography) rather than symptoms in all patients and was confirmed by biopsy in all patients. Overall, 56 patients were deceased, 36 (64%) due to metastasis. Of 279 patients who did not die or develop metastatic disease, 238 (85%) had at least 1 year of follow-up, 159 (57%) had at least 3 years of follow-up, and 67 (24%) had 5 or more years of follow-up.

GEP Class and Tumor Size

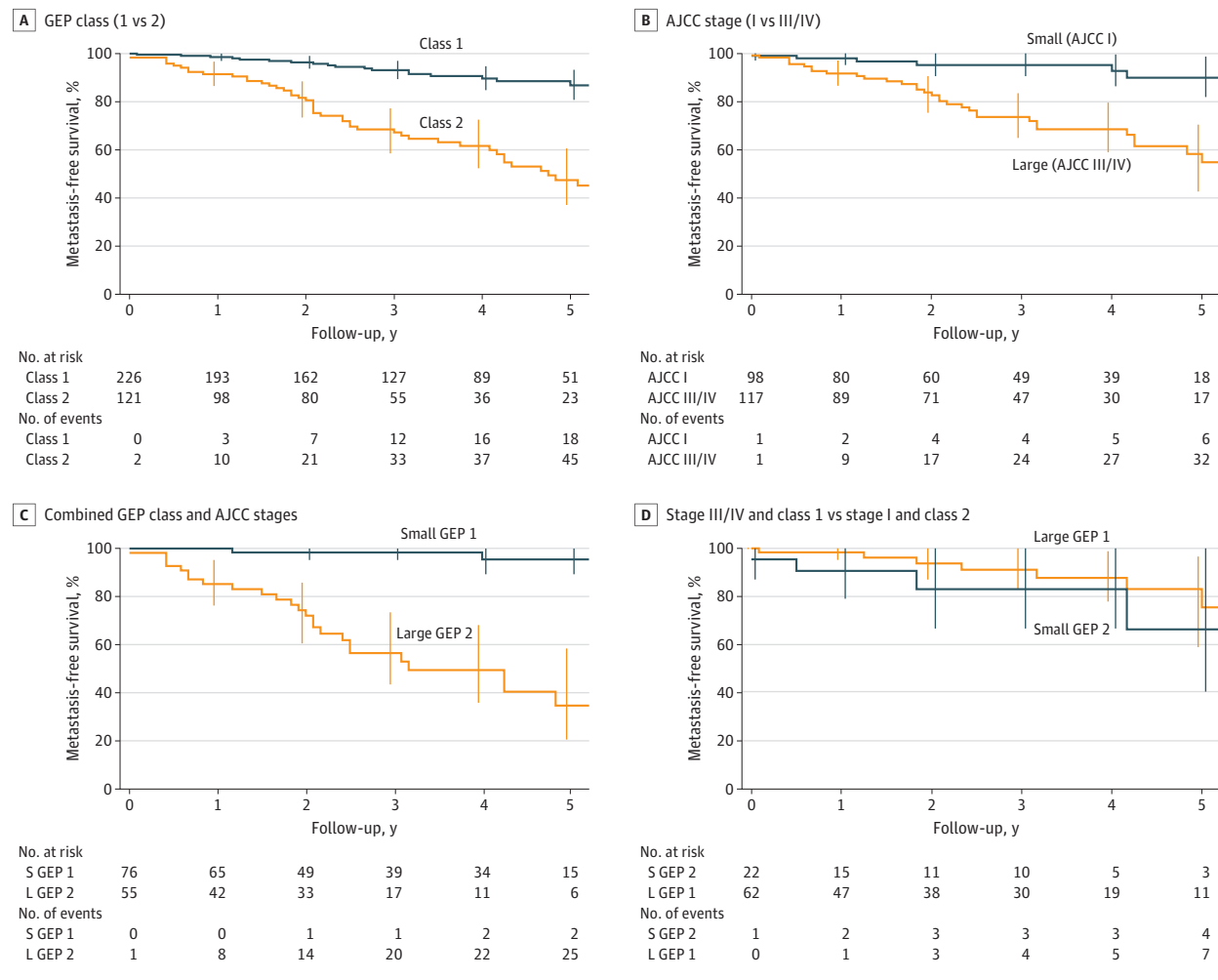
The number of patients with class 1 tumors who developed metastatic disease was too small for statistical analysis to

compare with those without metastatic disease. Patients with GEP class 2 tumors with (40 patients) and without metastasis (81 patients) were compared based on the eye involved, tumor location (choroid or ciliary body ± choroid), treatment type, COMS size, AJCC stage, and tumor size measured by tumor LBD and tumor thickness. Follow-up intervals were also compared between these groups. Within patients with class 2 tumors, those with metastasis had larger tumors compared with nonmetastatic tumors (COMS size: odds ratio, 1.8 [95% CI, 0.99-3.5]; $P = .06$; AJCC stage: odds ratio, 1.6 [95% CI, 1.02-2.5]; $P = .04$; larger mean LBD: difference, 1.7 [95% CI, 0.5-3.0] mm; $P = .01$; and greater mean thickness: odds ratio, 1.3 [95% CI, 1.04-1.5]; $P = .01$). No significant differences in other characteristics including follow-up length were observed (Table 2).

Observed MFS (3 and 5 Years)

The 3-year MFS was 93% (95% CI, 89%-97%) for patients with class 1 tumors and 67% (95% CI, 59%-77%) for those with class 2 tumors. The 5-year MFS was 87% (95% CI, 81%-93%) for patients with class 1 tumors and 47% (95% CI, 37%-61%) for those with class 2 tumors (Figure 1A). In general, the MFS was better for smaller (AJCC stage I) than larger tumors (AJCC stage III/IV) (Figure 1B). The MFS was best for GEP class 1 small tumors and worst for GEP class 2 large tumors (Figure 1C).

Figure 1. Observed Metastasis-Free Survival



Patients with small tumors with class 1 gene expression profiling (GEP) can be identified as a potentially curable. AJCC indicates American Joint Committee on Cancer; L, large; S, small.

However, there was no difference in the MFS between GEP class 1 large tumors and class 2 small tumors (Figure 1D). Increasing tumor size (thickness and LBD) was significantly associated with increased hazard ratio (1.16 [95% CI, 1.06-1.27]; $P < .001$ and 1.10 [95% CI, 1.02-1.19]; $P = .02$, respectively) of metastasis (eTable 1 in the Supplement).

Published MFS (3 and 5 Years)

The published MFS was obtained by performing a meta-analysis of data from published series (eTable 2 in the Supplement). The study by Walter et al²² reported only 5-year progression-free survival and did not include 3-year survival data, whereas Demirci et al¹⁶ reported only 3-year MFS but not 5-year data. These 2 studies did not meet the inclusion criteria and therefore were excluded. Heterogeneity was assessed directly and found to be nonsignificant, so fixed-effect estimates are presented. The MFS for patients with class 2 tumors in the published studies was 62% (95% CI, 57%-66%) and 40% (95% CI, 34%-46%) for 3 and 5 years, respectively (Figure 2B).

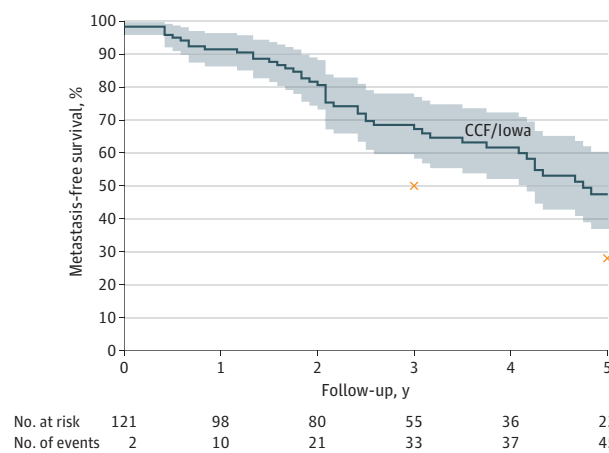
Predicted MFS

The predicted MFS for each patient as extracted from the GEP report was 50% and 28%, respectively (Figure 2). Therefore, we were only able to depict the predicted MFS as data points without 95% CI (limits) at 3 and 5 years (Figure 2). The predicted MFS was outside the 95% CI (limits) of the observed MFS (Figure 2), published MFS (meta-analysis), and each individual published study (except one⁶) (Figure 3).

Discussion

The observed MFS in our study of 67% (95% CI, 59%-77%) and 47% (95% CI, 37%-61%) at 3 and 5 years, respectively, was not different from those obtained by meta-analysis of the published studies for patients with class 2 tumors (62% [95% CI, 57%-66%] and 40% [95% CI, 34%-46%], respectively) (Figure 2B). The predicted MFS was below the 95% CIs (limits) of the observed MFS with the magnitude of the differences being

Figure 2. Observed and Predicted Metastasis-Free Survival



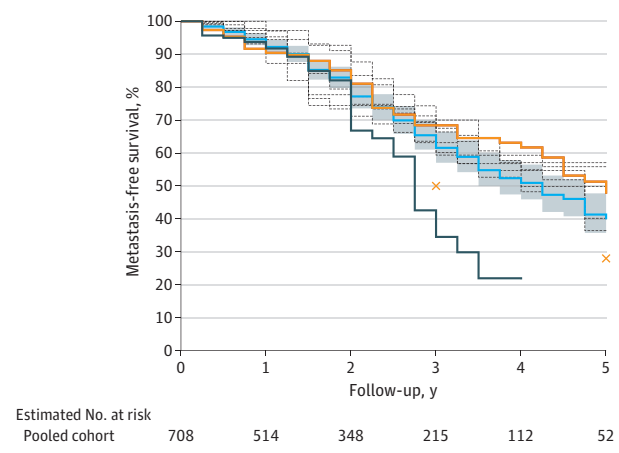
The solid line shows the observed metastasis-free survival with 95% CI (shaded). The predicted metastasis-free survival at 3 and 5 years are indicated as orange X marks. CCF/Iowa indicates Cleveland Clinic/University of Iowa.

similar at 3 and 5 years (17% and 19%, respectively). By inspection of the plotted MFS (and their 95% CI), it was striking to note overlap between our study and with each published study (except one⁶), suggesting the similarity in the observed and published MFS in all studies (Figure 3).

To our knowledge, the only study with worse MFS than all other studies was the GEP validation study by the COOG, which provides most of the data for the predicted MFS (DecisionDx-UM).⁶ Possible reasons for these differences are discussed below. Among the COOG study participants, 446 of 459 could be successfully classified by the GEP assay. Of 170 patients with class 2 tumor, metastasis was observed in 44 patients (26%) during the median study follow-up of 17.4 months (mean, 18.0 months). Data from an additional 68 cases with most recent censor date of June 9, 2011, are also included in the predicted MFS. The predicted MFS, reported as 50% and 28% at 3 and 5 years, respectively, could only be represented as data points without 95% CI (limits) as the full source data could not be found either in the published study or any publicly available data sets.⁶

Although our cohort is comparable with the COOG Study in regards to the number of metastatic events among patients with class 2 tumors (40 and 44, respectively), there were significant differences between the study populations regarding age (>66 years), tumor location (ciliary body involvement), and height (>6.2 mm) with a greater proportion of older patients, tumors involving the ciliary body, and thicker tumors in the COOG study (eTable 3 in the Supplement).⁶ Increasing age,¹⁷ ciliary body location,²³ and tumor size²³⁻²⁵ are well recognized to be independent poor prognostic factors.²⁶ The potential impact of tumor size on the risk for metastasis in patients with class 2 tumors is further supported by the fact that those with metastasis had larger tumors than those without metastasis (Table 2). Moreover, increasing tumor size (LBD, thickness) was associated with increased hazard ratio of metastasis. The potential impact of tumor size on MFS of class 2

Figure 3. Published (Meta-Analysis) Metastasis-Free Survival



The light blue line shows the meta-analysis-estimated metastasis-free survival with 95% CI (shaded). Each dotted line represents a study included in the meta-analysis. The observed metastasis-free survival (Cleveland Clinic/University of Iowa cohort) is indicated in orange. Observed metastasis-free survival, published metastasis-free survival (meta-analysis), and each individual published study (except Onken et al,⁶ show in dark blue) show better survival than the predicted metastasis-free survival.

tumors can also be observed by comparing MFS (at 3 years) of 84% and 50% in small and large tumors (AJCC stage I vs III/IV), respectively (Figure 1C and D). Notably, small class 2 tumors seem to have similar MFS (at 3 years and 5 years) as large class 1 tumors (Figure 1D), validating a previous observation.^{22,27} The 95% CIs are wide and overlap between the groups as tumors with such profiles are uncommon (22 [6%] and 62 [18%], respectively) and because of known correlation between the adverse GEP class and larger tumor size.^{28,29} The differences between the predicted MFS and the observed MFS were least for larger tumors (AJCC stage III/IV; 5% and 7%, respectively) and greatest for the small tumors (stage I; 34% and 40%, respectively) with near normal survival in small tumors with class 1 GEP. Patients with small tumors with class 1 GEP carry low metastatic risk and therefore can be identified as potentially curable.^{30,31} Taken together, these observations support the view that tumor size is an important factor that accounts for the differences between the predicted and the observed MFS.

Conceptually, our results unify, refine, and extend observations that define role of tumor size within the framework of GEP class; tumor size is associated with GEP class, GEP prognostic discrimination (class 1 and class 2), and survival in class 2 tumors. We believe that incorporation of tumor size into the GEP prediction algorithm will improve on its patient specific prediction.^{10,16,22,27,32}

Limitations and Strengths

One of the limitations of our study is its retrospective nature. We included consecutive patients to minimize selection bias and the patient follow-up protocols were standardized. The strengths of our study are the presence of 2-center consecutive patient independent data sets, with standardized methods for collecting metastasis and mortality data.^{7,33} All

patients in our study with metastatic disease had their metastases detected by surveillance imaging (100%) rather than by clinical symptoms and confirmed by biopsy (99%). None were lost to follow-up.

The ultimate purpose of prognostication is to identify at-risk individuals who could be offered adjuvant therapy with the aim of reducing tumor-specific mortality³⁴ as there is evidence that with patients with class 2 tumors could benefit from such treatments.³¹ Our study results may have bearing on the design of future adjuvant therapy trials as the use of predicted MFS in the absence of a control arm could be misleading. Any improvement in MFS over the predicted MFS by GEP (Decision Dx-UM) in a nonrandomized (treatment only) adjuvant therapy trial could be misinterpreted as a therapeutic

benefit, particularly in small tumors. Our data highlights the importance of considering both clinical and molecular data when designing and assessing outcomes in adjuvant therapy trials for uveal melanoma, particularly the need for a control arm (balanced or randomized).

Conclusions

The predicted MFS for metastatic tumors (class 2) may be too pessimistic than that observed by us and reported by others. Incorporation of tumor size in the prediction model may enhance its accuracy. Adjuvant therapy trials may not be able to rely confidently on predicted MFS to calculate efficacy.

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Supervision: Singh.

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